Exploratory Data Analysis on Used Cars Dataset

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Dataset Quality and Cleaning Metrics

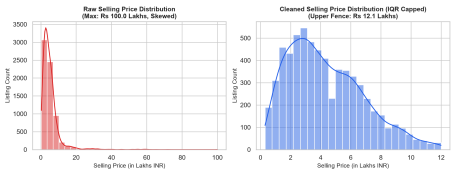
Source Dataset:	**Kaggle CarDekho Used Cars v3**
Raw Ingested Records:	**8,128 listings (13 attributes)**
Duplicate Rows Dropped:	**1,202 listings (retained 6,926)**
IQR Outliers Removed:	**328 Price • 167 Mileage**
Clean Analytics Sample:	**6,431 listings (92.9% retained)**

Summary of Core Business Findings

Depreciation Driver:	**Steepest drop during first 50,000 km**
Top Residual Value Brands:	**Toyota (Rs 7.1L) & Mahindra (Rs 6.5L)**
Fuel Premium:	**Diesel commands 38% premium over Petrol**
Transmission Premium:	**Automatic retains noticeable markup**
Primary Price Predictors:	**Horsepower (r=0.68) & Engine CC (r=0.53)**

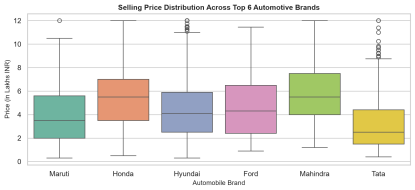
SIX VISUALIZATIONS ACROSS FIVE CHART TYPES WITH OBSERVATIONS

1. Price Distribution (Hist/KDE)



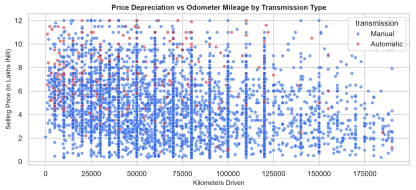
Raw prices are right skewed with extreme luxury tails. IQR filtering sets the upper fence at Rs 12.08 Lakhs, focusing the analysis on central consumer volume.

2. Top 6 Brands (Boxplot)



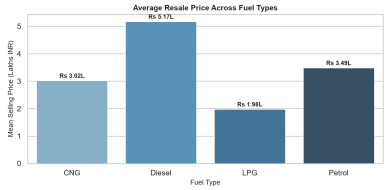
Toyota and Mahindra command the highest median prices due to durable SUV demand. Maruti and Hyundai dominate mass market sales volume with tight interquartile spreads.

3. Km vs Price (Scatter Plot)



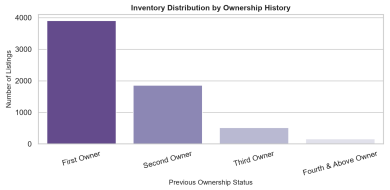
Clear non linear price decay as mileage climbs. Automatic cars maintain a steady price markup across all odometer readings compared to manual equivalents.

4. Fuel Type Resale (Bar Chart)



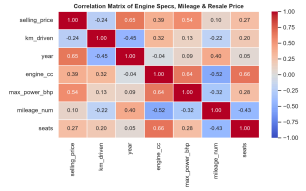
Diesel models average Rs 5.51 Lakhs, outpacing Petrol (Rs 3.99 Lakhs). CNG and LPG trade at substantial discounts due to taxi fleet usage and lower power output.

5. Ownership Status (Countplot)



First owners account for over 65% of listings, second owners represent 27%, and third or higher owners drop below 8%, showing rapid market drop-off with multiple transfers.

6. Correlation Matrix (Heatmap)



Max power (r=0.68) and engine displacement (r=0.53) are the strongest valuation drivers. Fuel efficiency shows an inverse trade-off with engine capacity (r=-0.58).